

Q1:JUN 2014 P2

4 (a) Define

(i) *streamline flow*,

(ii) *non-viscous flow*.

[2]

- (b) Fig. 4.1 shows a pipe carrying a fluid of density $1\,000\text{ kgm}^{-3}$. The diameter of end A is 6 cm while that of end B is 2 cm. At A the fluid is flowing at 2 m/s and has a pressure of 180 kPa.

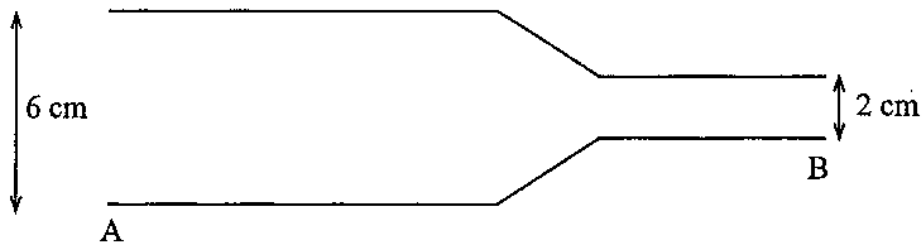


Fig. 4.1

Calculate

(i) the velocity at end B,

velocity = _____

(ii) the pressure at end B.

pressure = _____

[4]

Q2: NOV 2009 P2

- 8 (a) The flow of a fluid can be represented by $R = Av$,

where A is cross-sectional area and v is the speed of flow of the liquid.

- (i) Determine the base units of R .

base units = _____

- (ii) Hence identify the quantity R .

[3]

- (b) Bernoulli's equation $P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$, suggests that the term $\frac{1}{2}\rho v^2$ has the same base units as P .

Confirm the assertion.

[2]

Q3:JUN 2012 P2

- 5 (a) Explain the term *non-viscous fluid*.

[2]

- (b) Air flows over the upper surface of the wings of an aeroplane at a speed of 120 ms^{-1} and past the lower surfaces of the wings at 105 ms^{-1} . The total wing area of the plane is 25 m^2 and the density of air is 1.29 kgm^{-3} .

Calculate

- (i) the difference in pressure, Δp , between the upper and lower surfaces of the wings,

$$\Delta p = \underline{\hspace{2cm}}$$

- (ii) the 'lift' force, f , on the aeroplane.

$$f = \underline{\hspace{2cm}}$$

[4]

- (c) Explain why wings are shaped like an aerofoil.

[2]

Q4:NOV 2013 P2

- 4 (a) State **two** conditions necessary for the Bernoulli effect.

1. _____

2. _____

[2]

- (b) Fig. 4.1 shows a flowing fluid and an aerofoil.

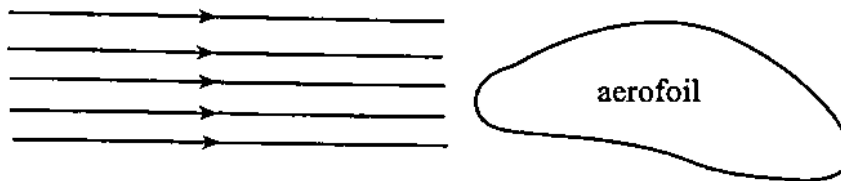


Fig. 4.1

- (i) On Fig. 4.1,
1. show the **two** vertical forces acting on the aerofoil,
 2. complete the fluid lines to illustrate the streamline flow of a fluid round an aerofoil.
- (ii) State with a reason whether the speed of the fluid is constant.

[5]

Q5:JUN 2019 P2

- 4 (b) Fig. 5.1 shows a horizontal pipe with a fluid, of density ρ , in motion. Part L has a diameter of 140 mm and part M a diameter of 90 mm. The fluid has a velocity of v_1 and v_2 at L and M respectively.

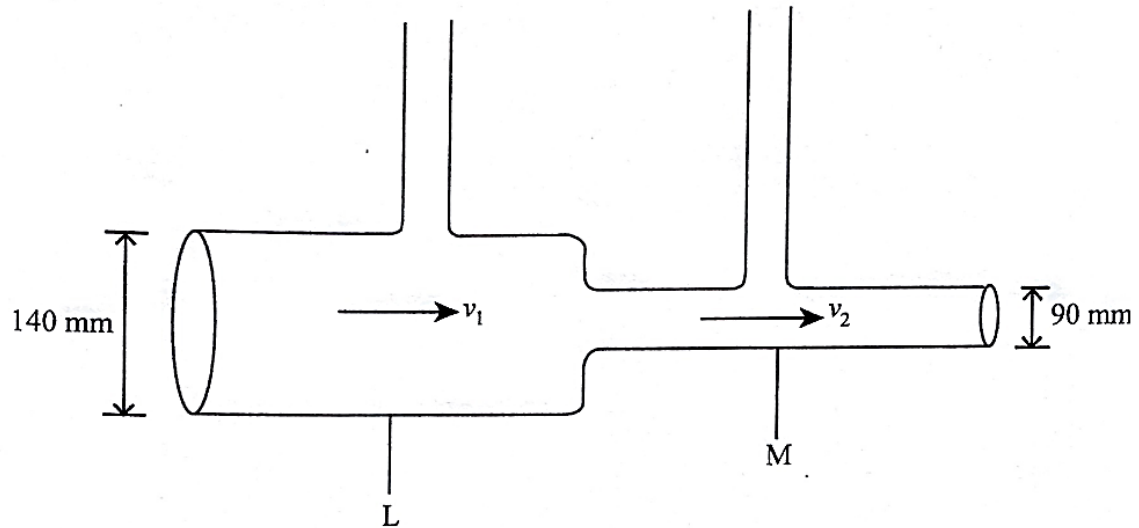


Fig. 5.1

- (i) On Fig. 5.1 show the vertical height of the fluid at L and M.
- (ii) Show that the pressure difference, Δp , at L and M is given by

$$\Delta p = \frac{1}{2} \rho v_1^2 \left(\frac{A_1^2}{A_2^2} - 1 \right).$$

- (iii) Calculate the pressure difference of a fluid in motion of density 890 kg m^{-3} when its velocity is 13 ms^{-1} .

Q6:NOV 2008 P5

- 1 (c) (i) Explain what is meant by lamina flow.
- (ii) Oil of density 0.800 g/cm^3 is flowing through a horizontal pipe as shown in Fig. 1.3.

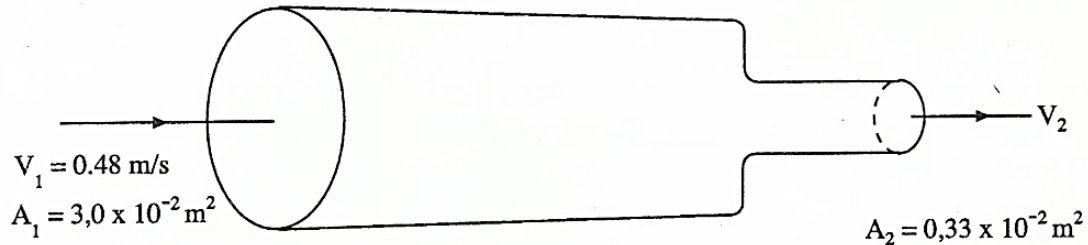


Fig. 1.3

Calculate the speed, V_2 , of the oil in the narrow pipe.

- (iii) An aerofoil shaped glass object is falling in calm air. A student places soft material vertically below the glass object on the ground, as shown in Fig. 1.4, to prevent it from breaking.

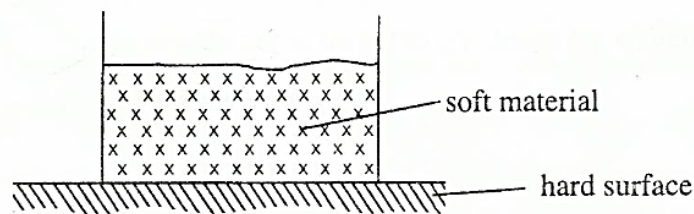
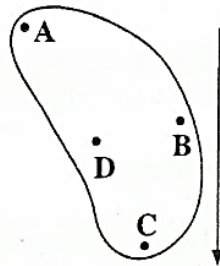


Fig. 1.4

Explain whether the student is likely to save the object from breaking.

[8]

Q7: NOV 2010 P5

- 2 (a) (i) State the Bernoulli effect.
- (ii) Give the conditions under which the Bernoulli effect is valid.
- (iii) With the aid of a labelled diagram, explain how the Bernoulli effect is applied in atomizers.

[8]

- (b) Fig. 2.1 shows an ideal incompressible fluid of density 790 kg m^{-3} flowing from A to B in a tube of varying cross-section. The cross-sectional area decreases from 390 mm^2 at A to 210 mm^2 at B. The fluid has a speed of $5.2 \times 10^{-3} \text{ m s}^{-1}$ at A.

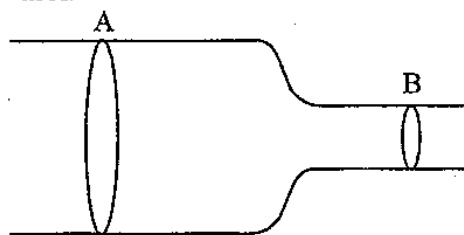


Fig 2.1

Calculate

- the speed of the fluid at B,
- the pressure difference between A and B.

[4]

Q8: JUN 2011 P5

- 3 (a) State what is meant by *an ideal fluid*. [2]
- (b) Derive the equation of continuity for an ideal fluid and state the principle on which it is based. [4]
- (c) (i) State the Bernoulli principle.
- (ii) A Bunsen burner shown in Fig. 3.1 is commonly used in school science laboratories for heating.

6.6 NON-VISCOUS FLUID FLOW P2, P3&P5 QUESTIONS

Compiled by: R. Nuga

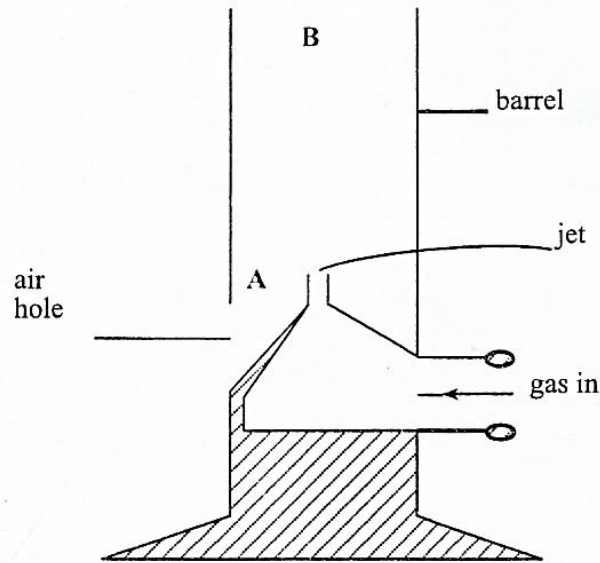


Fig. 3.1

Explain how the Bernoulli effect is applied in the structure of the burner.

[3]

- (iii) The cross-sectional area of the jet is $7.85 \times 10^{-7} \text{ m}^2$ and the velocity of the gas at that point is 12.5 m/s. The cross-sectional area of the barrel at B is $1.18 \times 10^{-4} \text{ m}^2$.

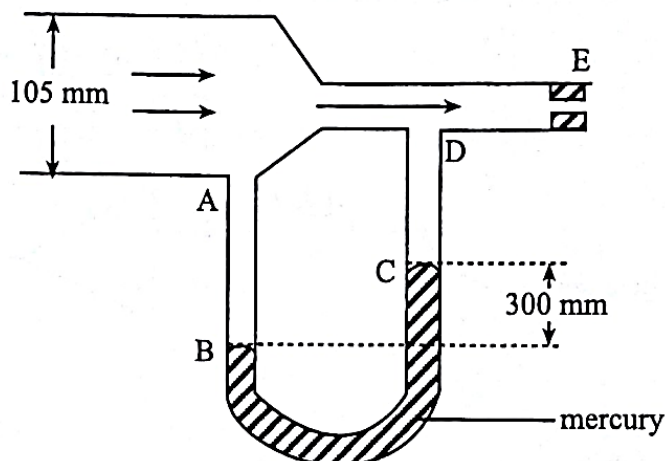
If the gas behaves as an ideal fluid, determine the pressure difference between points A and B. The density of the gas is 1.6 kg m^{-3} and the atmospheric pressure is $1.01 \times 10^5 \text{ Pa}$.

[3]

Q9:JUN 2013 P5

- 4 (a) State the equation of continuity, with terms explained, for the flow of an ideal fluid.
- (b) (i) Fig. 4.1 shows a U-tube manometer connected to two different parts of a pipe with flowing water.

[2]



6.6 NON-VISCOUS FLUID FLOW P2, P3&P5 QUESTIONS

Compiled by: R. Nuga

1. Explain why the mercury levels are as shown in the diagram.
2. The speed of water in the wider part is 10 m/s and the densities of water and mercury are 1000 kgm^{-3} and $13\,600 \text{ kgm}^{-3}$ respectively.

If the water is assumed to behave ideally, determine the diameter of the narrow part.

[8]

- (ii) The end E of the tube in Fig. 4.1 can be adjusted to further reduce the cross-sectional area at this part.

Suggest and explain **one** advantage of such a system for irrigation purposes.

[2]

Q10:NOV 2014 P5

- (c) (i) State what is meant by lamina flow.
- (ii) Fig. 1.2 shows water coming out of a tap.

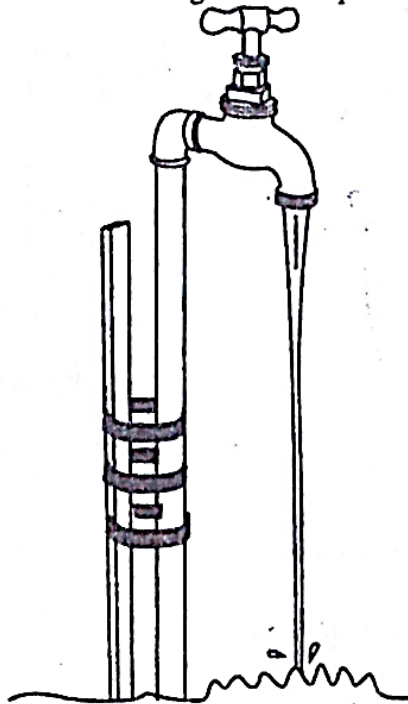


Fig. 1.2

Explain why the water thins out as it flows to the ground.

[4]

Q11:JUN 2015 P5

- 4 (a) (i) Explain what is meant by
1. *lamina flow*,
 2. *the Bernoulli effect*.
- (ii) State the two conditions considered in deriving the Bernoulli equation. [4]
- (b) Fig. 4.1 shows oil of density 800 kg m^{-3} flowing through a horizontal tube.

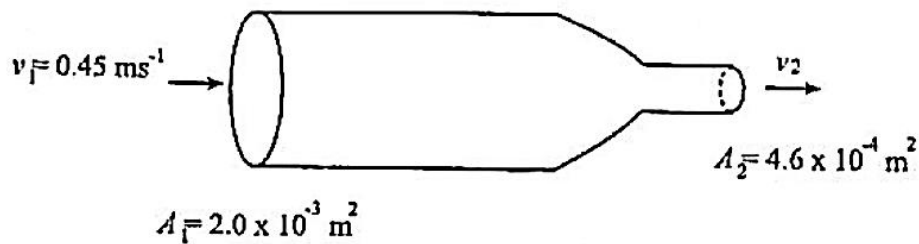


Fig. 4.1

Calculate

1. velocity, v_2 .
 2. the pressure difference between the two ends of the tube. [4]
- (c) As the oil flows in the tube state a quantity that is
- (i) conserved,
 - (ii) not conserved. [2]
- (d) Suggest why it is safer to sail a boat across a wider part of a river than its narrow part. [2]

Q12:JUN 2017 P5

- 1 (d) (i) Explain the following terms as applied to the motion of an ideal fluid:
1. *steady flow*,
 2. *incompressible flow*,
 3. *non-viscous flow*.

- (ii) 1. State the principle on which the equation of continuity for the flow of an ideal fluid is based.
2. Fig. 1.1 shows a water hose P and an end fitting Q of internal diameters 18.0 mm and 4.0 mm respectively.

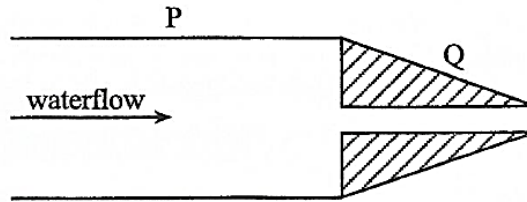


Fig. 1.1

Given that water emerges from Q with speed 4.3 ms^{-1} , calculate the speed of water in the hose.

[7]

Q13:JUN 2018 P5

- 3 (a) Explain what is meant by an ideal fluid. [2]
- (b) Fig.3.1 shows a venturi tube used to measure the speed of water v_1 in a pipe.

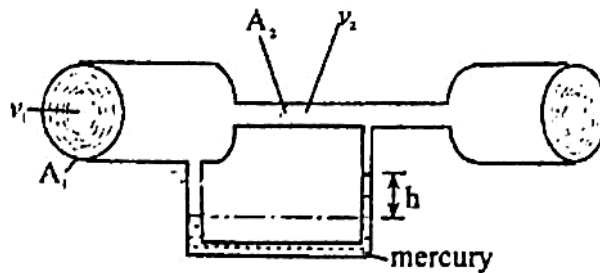


Fig.3.1

The cross sectional areas at a wide and narrow sections are A_1 and A_2 where $A_2 = \frac{A_1}{4}$. The height $h = 25 \text{ mm}$, density of mercury $\rho_{\text{Hg}} = 13.600 \text{ kgm}^{-3}$ and density of water $\rho_w = 1000 \text{ kgm}^{-3}$.

- (i) Express v_2 in terms of v_1 .
- (ii) Calculate the values of
1. v_1 ,
 2. v_2 .

[7]

6.6 NON-VISCOUS FLUID FLOW P2, P3&P5 QUESTIONS

Compiled by: R. Nuga

- (c) The window and the door of a room are both completely open. The door opens inwards.

Explain why the door tends to shut with a bang when only slightly open. [3]

Q14:NOV 2006 P3

- 6 (c) (i) State the Bernoulli principle. [6]
- (ii) Explain how the Bernoulli effect is applied in a filter pump. [4]
- (d) Compare the advantages and disadvantages of a thermocouple and a resistance thermometer. [4]

Q15:JUN 2019 P3

- 4 (c) (i) Explain what is meant by
1. *incompressible flow*,
 2. *non-viscous flow of an ideal fluid*.
- (ii) An ideal fluid is in a steady flow from a wider to a narrower section of a horizontal pipe.
1. State the section in which pressure is higher.
 2. Show that its speed of flow increases.
 3. Calculate the difference in pressure, given that the fluid has a density of 800 kg m^{-3} and is flowing with velocities 0.74 ms^{-1} and 3.5 ms^{-1} in the wider and narrower sections respectively.
- [7]

Q16:NOV 2016 P5

- 2 (a) Explain the term
- (i) *laminar flow*,
 - (ii) *incompressible flow*.
- [2]

6.6 NON-VISCOUS FLUID FLOW P2, P3&P5 QUESTIONS

Compiled by: R. Nuzo

- (b) (i) Derive the equation of continuity for the flow of an ideal fluid.
- (ii) Fig. 2.1 shows oil of density 800 kg/m^3 flowing in a tube.

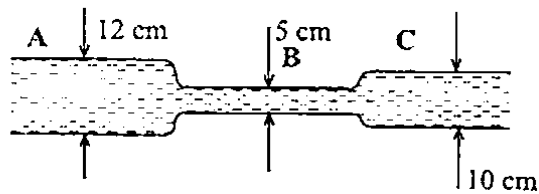


Fig. 2.1

The flow rate of oil at C is $5 \text{ cm}^3/\text{s}$. Find the velocity of the oil in section A and B.

- (iii) The pressure in Section A in Fig. 2.1 is 100 kPa.
Determine the pressure in B.
- (iv) Suggest any deduction that could be made from the answer in b(iii).

[10]